

Heart Disease Prediction Using Machine Learning

Binary Classification on CDC BRFSS 2020 Survey Data

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ABSTRACT

Heart disease is the leading cause of death globally, claiming approximately 17.9 million lives per year — 31% of all deaths (World Health Organization, 2023). Early identification of high-risk individuals is a well-established clinical priority, yet most patients are not diagnosed until after a serious cardiac event. This study applies three binary classification algorithms — Logistic Regression, Random Forest, and Support Vector Machine — to a representative 2,059-patient sample drawn from the U.S. CDC Behavioral Risk Factor Surveillance System 2020 survey. The sample preserves the real-world disease prevalence of 8.6%, and class imbalance is addressed through `class_weight='balanced'` across all models. Hyperparameters are selected via GridSearchCV with 5-fold stratified cross-validation, scored on F1. Random Forest achieves the highest AUC-ROC (0.855) and Logistic Regression the highest disease recall (80%). Feature importance analysis identifies age category, general health status, and difficulty walking as the three strongest predictors. Ethical considerations including demographic bias, self-reported data limitations, and the asymmetric cost of false negatives are discussed.

1. Introduction

Cardiovascular disease (CVD) remains the world's most significant cause of preventable death. Coronary heart disease — the blockage of arteries supplying the heart — accounts for the majority of CVD fatalities. A central challenge in clinical care is that patients typically receive a diagnosis only after a heart attack or stroke, at which point tissue damage has already occurred. Earlier identification of at-risk individuals allows for lifestyle intervention, medication, and monitoring that can meaningfully reduce the probability of a serious event.

Machine learning offers a path toward population-level risk stratification using data that already exists. The CDC's Behavioral Risk Factor Surveillance System (BRFSS) collects self-reported health and lifestyle data from over 400,000 U.S. adults annually. The 17 questions used in this study — covering age, smoking, diabetes, physical activity, and general health, among others — require no clinical test, no blood draw, and no specialist appointment. A model that can identify high-risk patients from this data alone represents a meaningful step toward scalable preventive screening.

This project builds, tunes, and compares three classification algorithms on a 2,059-patient sample from the CDC BRFSS 2020 dataset. The objectives are: (1) implement Logistic Regression, Random Forest, and SVM with systematic hyperparameter optimisation; (2) handle the real-world class imbalance correctly without distorting the data distribution; (3) interpret which features drive

predictions; and (4) critically assess the ethical implications of automated health screening.

2. Methodology

2.1 Dataset

The dataset is the CDC Behavioral Risk Factor Surveillance System 2020, made available on Kaggle by Kamilpytlak (2021). The full file contains 319,795 responses and 18 features. The target variable — HeartDisease — indicates whether the respondent has ever been diagnosed with coronary heart disease or myocardial infarction. The full dataset carries an 8.56% positive rate, reflecting real U.S. population prevalence.

A representative sample of **2,059 patients** was drawn: 177 with heart disease (8.6%) and 1,882 without (91.4%). This is a deliberate methodological choice. A common shortcut is to force a 50/50 class split, which makes models appear more balanced but fundamentally misrepresents how rare the condition is in the real world — and therefore how a deployed model would actually perform. Preserving the natural distribution and handling imbalance through model-level weighting is both more honest and more clinically meaningful.

Feature	Description	Encoding
BMI	Body Mass Index	Continuous
AgeCategory	Age band: 18–24 up to 80+	Ordinal (0–12)
GenHealth	Self-rated health: Poor to Excellent	Ordinal (0–4)
Diabetic	Diabetes status	Ordinal (0–2)
PhysicalHealth	Poor physical health days (last 30)	Continuous
MentalHealth	Poor mental health days (last 30)	Continuous
SleepTime	Average sleep hours	Continuous
Sex	Male or Female	Binary (0/1)
Smoking	Smoked 100+ cigarettes lifetime	Binary (0/1)
AlcoholDrinking	Heavy drinker	Binary (0/1)
Stroke	Ever had a stroke	Binary (0/1)
DiffWalking	Difficulty walking or climbing stairs	Binary (0/1)
PhysicalActivity	Exercise in past 30 days	Binary (0/1)
Asthma	Ever had asthma	Binary (0/1)
KidneyDisease	Ever had kidney disease	Binary (0/1)
SkinCancer	Ever had skin cancer	Binary (0/1)
Race	White / Black / Asian / Hispanic / Other	One-hot (5 cols)
HeartDisease	Ever diagnosed with CHD or MI — TARGET	Binary (0/1)

Table 1: All 17 input features, descriptions, and encoding strategy.

2.2 Data Preprocessing

Yes/No columns were encoded as 0/1. Sex was encoded Male=1, Female=0. AgeCategory was mapped to an ordinal scale from 0 (18–24) to 12 (80 or older), preserving the natural age ordering. GenHealth was similarly mapped 0 (Poor) to 4 (Excellent). Diabetic was treated as ordinal: No=0, Borderline/Gestational=1, Yes=2. Race was one-hot encoded with the first category dropped to avoid multicollinearity, producing five additional binary columns. After encoding, the dataset has 21 input features and zero missing values.

The data was split 80/20 into training (1,647 patients) and test (412 patients) sets using stratified sampling, preserving the 8.6% disease rate in both subsets. All features were scaled using StandardScaler, fitted exclusively on the training set. Fitting the scaler on all data before the split would constitute data leakage — the test set would influence the scaling parameters, producing falsely optimistic evaluation results.

2.3 Handling Class Imbalance

With 91.4% of patients disease-free, a naive classifier predicting 'no disease' for every patient achieves 91.4% accuracy while failing completely at the actual task. All three models were trained with `class_weight='balanced'`, which automatically adjusts the per-sample loss weight inversely proportional to class frequency. This penalises the model more heavily for misclassifying disease cases, without altering the underlying data distribution or overstating how common heart disease actually is.

2.4 Hyperparameter Tuning

GridSearchCV with 5-fold stratified cross-validation was applied to all three models, scoring on F1. F1 was chosen over accuracy because it balances precision and recall — both relevant in a medical context where false positives waste clinical resources and false negatives delay treatment. Stratified folds ensure each split preserves the 8.6% disease rate.

Model	Parameter Grid	Best Parameters
Logistic Regression	$C \in \{0.01, 0.1, 0.5, 1, 5, 10\}$	$C = 0.01$
Random Forest	$n_estimators \in \{100, 200, 300\}$, $max_depth \in \{5, 10, 20, None\}$, $min_samples_split \in \{2, 5, 10\}$	$n=100$, $depth=5$, $split=10$
SVM (RBF)	$C \in \{0.1, 1, 10, 100\}$, $gamma \in \{scale, auto, 0.01, 0.001\}$	$C=0.1$, $\gamma=0.001$

Table 2: GridSearchCV parameter grids and selected best parameters (F1 scoring, 5-fold CV).

3. Algorithm Selection

3.1 Logistic Regression

Logistic Regression models the log-odds of heart disease as a weighted linear combination of features, producing probability estimates through the sigmoid function. It was included as the interpretable baseline: each coefficient directly expresses the direction and magnitude of a feature's influence on predicted risk, making its decisions auditable in a clinical context. The optimal $C=0.01$ indicates strong L2 regularisation, appropriate given the limited number of positive cases in the training set (152 disease patients). Despite its simplicity, Logistic Regression achieves the highest

disease recall of the three models (Pedregosa et al., 2011).

3.2 Random Forest

Random Forest constructs an ensemble of decision trees, each trained on a random subset of the training data and features, then aggregates predictions by majority vote (Breiman, 2001). This reduces the variance of individual trees while retaining their ability to capture non-linear relationships. The constrained depth (max_depth=5) and elevated split threshold (min_samples_split=10) act as regularisers, preventing trees from memorising the minority class noise. A further advantage of Random Forest is its native provision of Gini-based feature importance scores, which allow identification of which variables contribute most to predictions across the full ensemble.

3.3 Support Vector Machine (RBF Kernel)

The SVM with RBF kernel identifies the maximum-margin hyperplane separating classes in a kernel-transformed feature space (Cortes and Vapnik, 1995). The soft-margin parameter C=0.1 permits misclassifications, appropriate here given the imbalance — tighter margins would force the boundary toward the numerous healthy patients. The small gamma value (0.001) produces a wide kernel with broad influence, reducing sensitivity to individual noisy samples. probability=True enables Platt-scaled probability outputs required for AUC-ROC computation.

4. Results and Analysis

4.1 Model Performance

Model	CV Accuracy	±Std	Test Accuracy	Disease Recall	F1	AUC-ROC
Logistic Regression	77.5%	±1.3%	76.5%	80.0%	0.366	0.849
SVM (RBF)	86.9%	±2.0%	84.2%	54.3%	0.369	0.828
Random Forest	82.6%	±1.6%	82.3%	74.3%	0.416	0.855 ★

Table 3: Performance on the 412-patient held-out test set. Recall = recall for the heart disease class.

Why SVM has 84% accuracy but only 54% recall: SVM learns to predict 'no disease' for borderline cases, exploiting the 91.4% majority class. Despite class_weight=balanced, it does not generalise the minority class as effectively as the other two models on this survey dataset. This demonstrates precisely why accuracy alone is a misleading metric for imbalanced medical classification tasks.

4.2 Model Selection

Random Forest is selected as the best overall model based on AUC-ROC (0.855), which measures discriminatory ability across all classification thresholds and is the most appropriate single metric for imbalanced binary classification (Fawcett, 2006). It also achieves the best balance of test accuracy (82.3%) and disease recall (74.3%). Logistic Regression achieves higher recall (80%) but lower AUC (0.849) and accuracy (76.5%), making it the preferred choice in contexts where missed diagnoses are clinically unacceptable.

4.3 Feature Importance (Random Forest)

Gini-based feature importance scores identify the five strongest predictors of heart disease in our CDC sample:

- **AgeCategory (28.7%).** The dominant predictor by a significant margin. Disease prevalence in our sample rises from 0.6% in the 18–24 group to 31.6% in patients aged 80 or older. This gradient is consistent with decades of cardiovascular epidemiology.
- **GenHealth (13.4%).** Self-reported general health serves as a powerful composite signal. Patients rating their health as 'Poor' had a 32.1% disease rate; those rating it 'Excellent' had a rate of 2.1%. One question captures the aggregate effect of all comorbidities.
- **DiffWalking (10.0%).** Difficulty walking or climbing stairs reflects physical deconditioning — a measurable downstream consequence of reduced cardiac output.
- **PhysicalHealth (8.6%).** The number of poor physical health days in the past month correlates with chronic illness burden and residual symptoms of cardiovascular disease.
- **Diabetic (8.2%).** Diabetes damages arterial endothelium through the same mechanisms as coronary artery disease. In our data, diabetic patients had an 18.6% disease rate versus 7.1% for non-diabetics.

4.4 Limitations

Several limitations constrain the scope of these findings. The sample of 177 disease patients provides a relatively small base for the models to learn from, and the 35 disease cases in the test set introduce statistical uncertainty in point metrics like recall. The data is entirely self-reported, introducing measurement noise. The sample reflects the U.S. population and may not generalise internationally. Finally, one-hot encoding of Race introduces features that may capture structural healthcare inequalities rather than biological risk.

5. Ethical Discussion

5.1 Bias and Fairness

The BRFSS dataset reflects a U.S. respondent population with known disparities in healthcare access, health literacy, and insurance coverage across racial and socioeconomic groups. Race is included as a feature, which risks encoding structural inequalities into the model's predictions rather than isolating biological risk. Any deployment must evaluate performance separately by subgroup — a model that appears accurate at the population level may still systematically under-screen specific groups (Barocas et al., 2019).

5.2 Self-Reported Data and Measurement Error

BRFSS relies on telephone surveys, introducing well-documented sources of measurement error: respondents may under-report stigmatised behaviours such as smoking and alcohol consumption, may be unaware of undiagnosed conditions, or may interpret questions inconsistently. This noise is part of why test accuracy is approximately 82% rather than the 88–92% typically reported on clinical measurement datasets. It is a feature of the data type, not a flaw in the modelling approach.

5.3 The Asymmetric Cost of False Negatives

A false negative — predicting 'no disease' for a patient who has it — could delay or prevent diagnosis and treatment. A false positive — flagging a healthy patient — causes unnecessary anxiety and

wastes clinical resources. These costs are not symmetric: in most screening contexts, missed diagnoses are more serious. This motivated using F1 as the tuning metric and `class_weight='balanced'` as the imbalance strategy. In a real deployment, the classification threshold would likely be lowered below 0.5 to increase recall further, at the cost of more false alarms.

5.4 Privacy and Regulatory Compliance

Any system applying these models to real patients processes protected health information subject to EU GDPR (right to explanation, data minimisation, consent) and U.S. HIPAA (access controls, audit trails, anonymisation). The BRFSS data used here is publicly anonymised, but a clinical deployment would require re-anonymisation of local patient records and role-based access controls (European Parliament, 2016).

5.5 Decision Support, Not Replacement

This model is a decision-support tool. The clinical decision always rests with the physician. Research on automation bias shows that clinicians can over-rely on algorithmic outputs (Cummings, 2004). To mitigate this, any deployed version should use SHAP (Lundberg and Lee, 2017) to provide patient-level explanations — for example: 'This patient was flagged primarily because they are in the 70–74 age group, report fair general health, and have difficulty walking.'

6. Conclusion

This project demonstrates that heart disease risk can be predicted from 17 self-reported lifestyle and health questions with meaningful accuracy. On a 2,059-patient sample from the CDC BRFSS 2020 survey at the real-world disease rate of 8.6%, Random Forest achieves the strongest overall performance — AUC-ROC of 0.855, test accuracy of 82.3%, and disease recall of 74.3% — after GridSearchCV tuning with `class_weight='balanced'`. Age, general health rating, and difficulty walking are the three most informative predictors, consistent with established cardiovascular risk literature.

Future directions include expanding the sample to improve minority class representation, evaluating fairness across demographic subgroups, integrating SHAP explainability, and validating on external clinical cohorts. Deployed responsibly as a decision-support tool, models of this kind could contribute to earlier cardiovascular risk identification at population scale.

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